

Appendix A - Description of test areas at Andøya



- RED** = Official Test Area 1
- Green** = Official Test Area 2
- Blue** = Official Test Area 3

Survey points

Notice: Geodetic reference frame for plane coordinates is EUREF89.

Differences between EUREF89 and WGS84, see Appendix H.

Seven significant decimal digits for latitude and longitude will ensure cm-precision.

*Table 1: Coordinates for Jammertest 2026 (Physical height = heights in The Norwegian NN2000 height system)
The heights are approximate until measurements are carried out in September*

Point-ID	UTM33-North	UTM33-East	Physical height	Latitude	Longitude	Ellipsoidal height	Mark
ORIG	7679455.500	536952.700	30.000	69.2225166	15.9334241	66.00	
A50	7679505.405	536955.446	30.000	69.2229637	15.9335127	66.00	
A100	7679555.311	536958.193	30.000	69.2234108	15.9336011	66.00	
A150	7679605.216	536960.939	30.000	69.2238580	15.9336898	66.00	
B50	7679428.169	536994.546	30.000	69.2222658	15.9344705	66.00	
B100	7679400.838	537036.392	30.000	69.2220150	15.9355167	66.00	
B150	7679373.507	537078.238	30.000	69.2217642	15.9365630	66.00	
C50	7679432.926	536908.108	30.000	69.2223203	15.9322893	66.00	
C100	7679410.352	536863.515	30.000	69.2221239	15.9311544	66.00	
C150	7679387.777	536818.923	30.000	69.2219276	15.9300194	66.00	
D50	7679478.074	536997.292	30.000	69.2227129	15.9345589	66.00	
D100	7679500.648	537041.885	30.000	69.2229092	15.9356941	66.00	
D150	7679523.223	537086.477	30.000	69.2231055	15.9368289	66.00	
E50	7679405.595	536949.954	30.000	69.2220694	15.9333354	66.00	
E100	7679355.689	536947.207	30.000	69.2216223	15.9332470	66.00	
E150	7679305.784	536944.461	30.000	69.2211752	15.9331584	66.00	
F50	7679482.831	536910.854	30.000	69.2227673	15.9323776	66.00	
F100	7679510.162	536869.008	30.000	69.2230181	15.9313314	66.00	
F150	7679537.493	536827.162	30.000	69.2232689	15.9302849	66.00	
Z211	7680321.98	537737.24	150.000	69.2301778	15.9535787	185.86	
Z212	7679941.11	537693.08	150.000	69.2267688	15.9523135	185.86	
Z213	7678454.29	537094.18	150.000	69.2135195	15.9366108	185.87	
Z214	7678230.54	536345.37	150.000	69.2116148	15.9176221	185.91	
Z216	7680389.12	537316.50	30.000	69.2308382	15.9429750	65.88	
Z218	7680161.50	537228.58	30.000	69.2288093	15.9406654	65.88	
Z219	7679984.08	537118.29	30.000	69.2272336	15.9378105	65.88	
Z220	7679582.01	536821.06	30.000	69.2236689	15.9301479	65.89	
Z221	7678711.68	536064.22	30.000	69.2159667	15.9107055	65.92	
Z222	7678780.32	536166.47	30.000	69.2165686	15.9133130	65.92	
Z223	7678938.97	536333.96	30.000	69.2179687	15.9176019	65.91	
Z224	7679134.73	536489.45	30.000	69.2197031	15.9216025	65.91	
301	7670668.615	530198.748	8.408	69.1445606	15.7600883	44.52	

302	7670680.369	530212.300	8.494	69.1446644	15.7604330	44.61
303	7677529.277	534164.531	8.165	69.2056106	15.8623208	44.16
304	7677547.576	534161.777	8.091	69.2057751	15.8622579	44.08
305	7678210.998	534007.445	6.021	69.2117434	15.8585974	42.02
306	7678203.811	534010.272	6.108	69.2116786	15.8586663	42.10
REFANT	7685369.617	538256.263	5.424	69.2753655	15.9687079	41.27

Description of Test Area 1 (maps from 2025)

Overview of survey points



More detailed view of surveyed points outside Bleik samfunnshus.

Description of Test Area 2

Test Area 2 is the parking area at the end of a dirt road. (Position N 69.2225°, Ø 15.9335°).

Most of the testing will be conducted at the parking or the surrounding area.

The setup is based around known positions, distances and controlled RF power levels for the tests.

Barracks and parking plan 2026

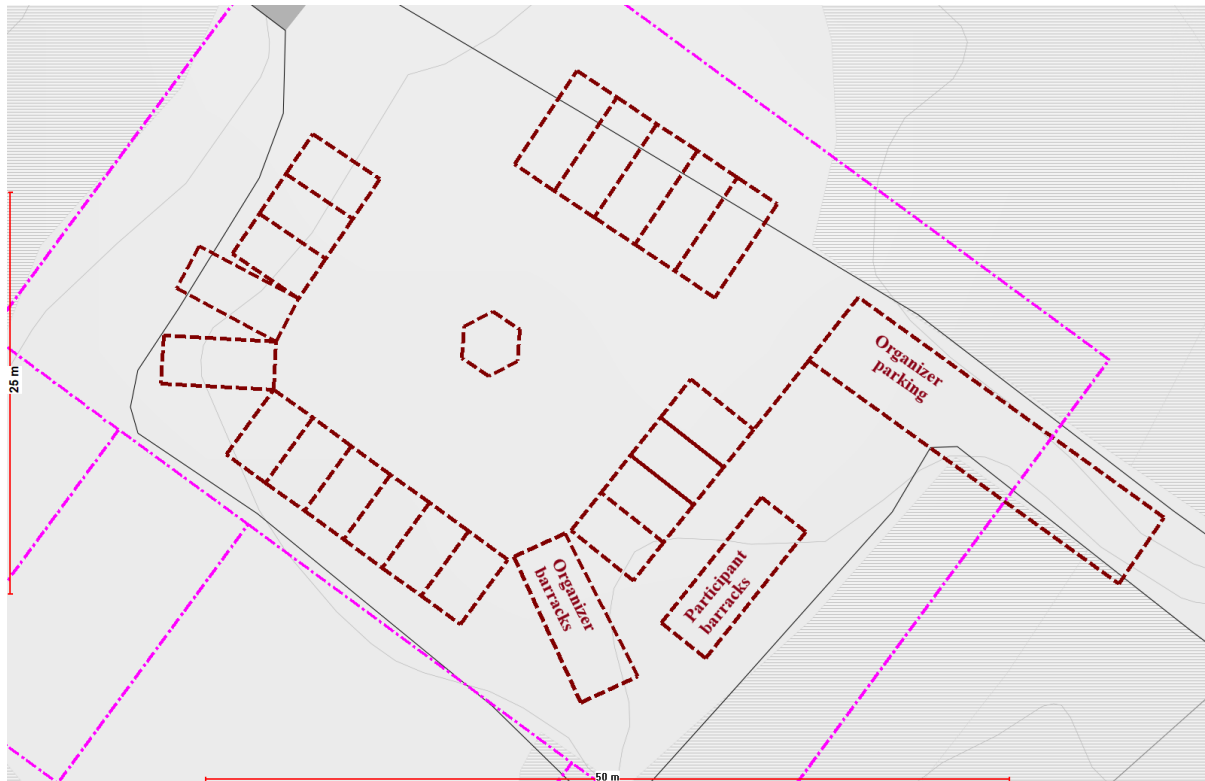


Figure 1: Barracks and parking plan 2026

Drone areas 2026

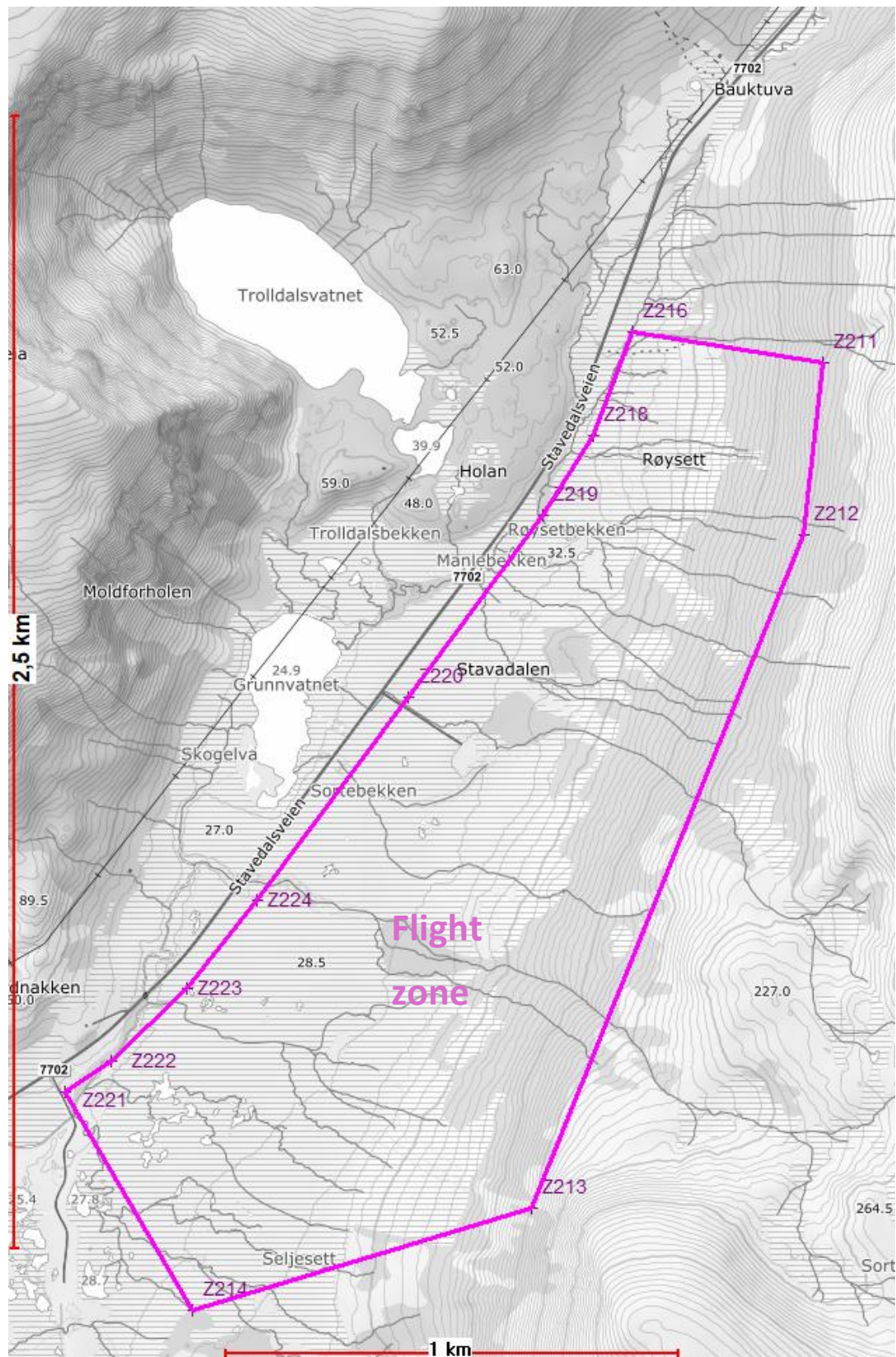


Figure 2: Drone area boundary

Jammer and drone markers 2026

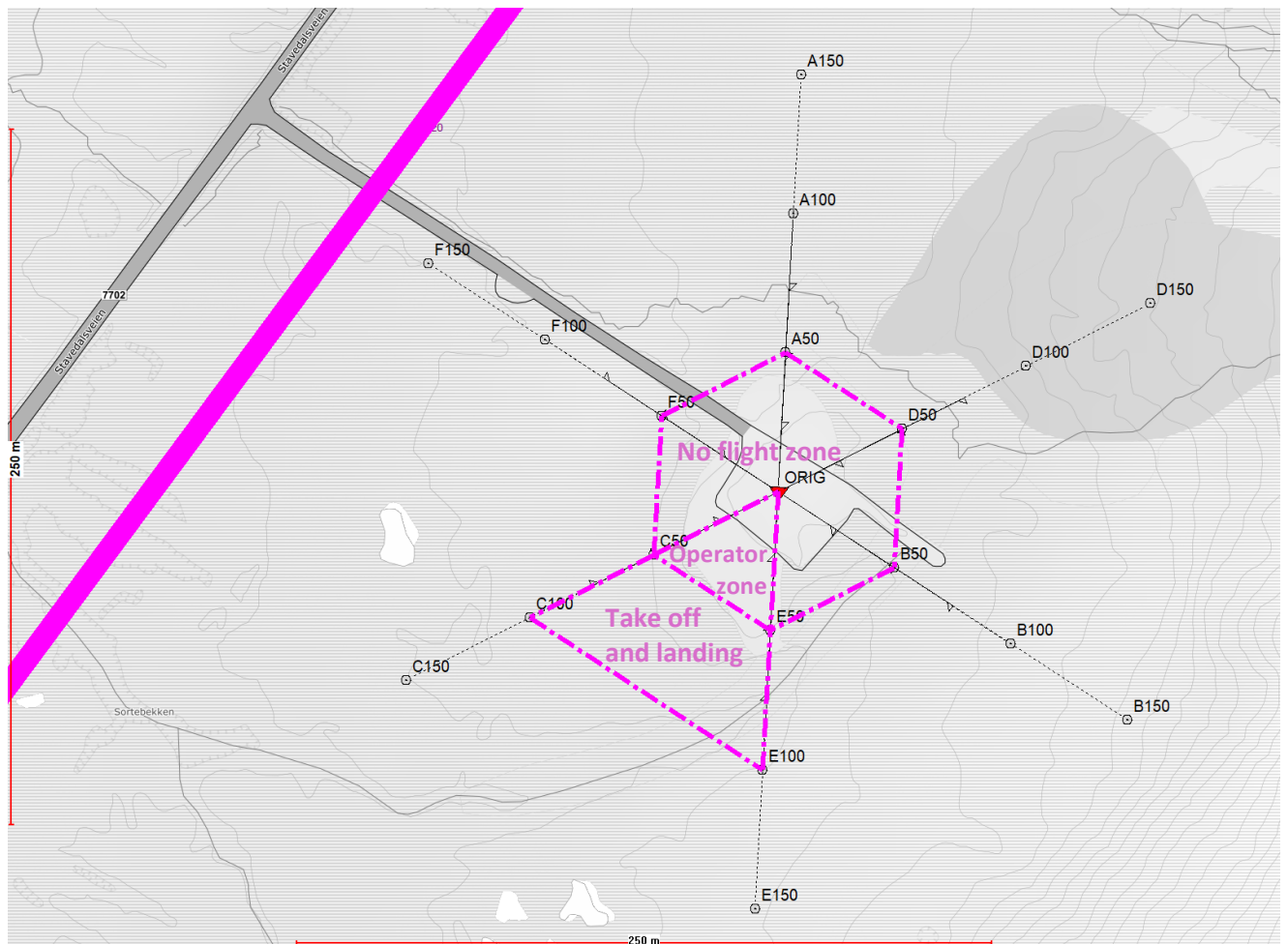


Figure 3 Test Area 2, directions for where we place the jammers and several new directions (2026) that will work as visual markers from the air as “drone waypoints”.

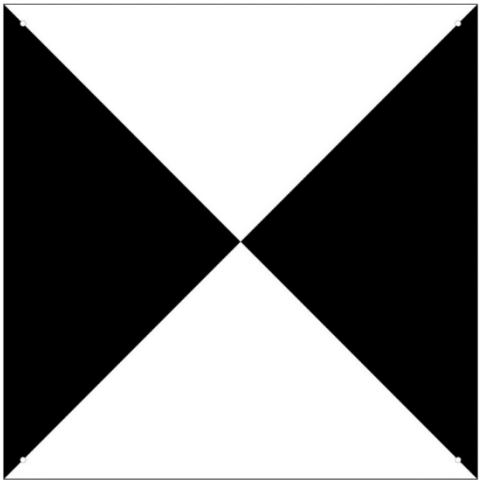
The test setup at location 2

A, B and C axis are positions used for placement of jammers, or spoofing equipment, as shown in figure 3. A, B and C axis is separated 120 degrees apart. Points will be marked with “wood sticks” (trelekter).

D, E and F axis are continuations of C, A and B axis. They will be marked with drone waypoints visual from the air.

D, E and F- axis will have black and white drone markers, which should be visible at 100 meters above ground level.

Example: drone marker/ visual waypoint. (Dimensions 50 cm x 50 cm)



Fiducial marks added to drone markers for Test Area 2.

Axis marked with color and distance marked with squares.

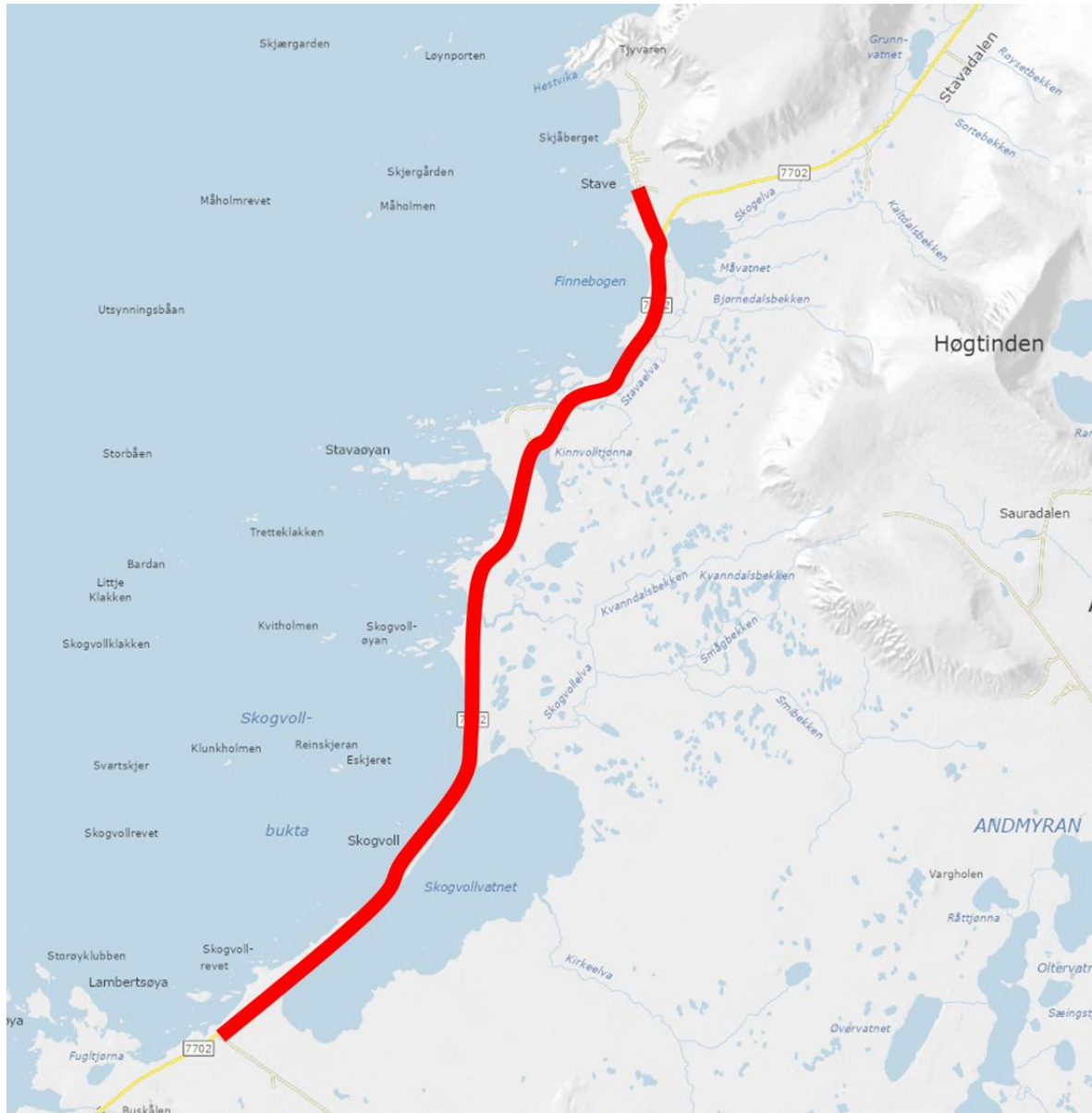
Red	=	D - direction
Blue	=	E - direction
Green	=	F - direction
1 square	=	50 meters
2 squares	=	100 meters
3 squares	=	150 meters

Example: D – direction @ 150 meters.



Description of motorcade route(s) on Andøya, Test Area 3 (2025)

The start is at Stave community house (69.212187 North, 15.858559 East), the small jammers can be used at the intersection between county road 7702 and communal road "Okleveien" (69.14409 North, 15.75847 East). The figure below shows the stretch that can be used for the motorcade (Red line).



The road is quite narrow 5.1 meters with a speed limit of 80 km/h. The traffic volume is low with about 1000 vehicles per day. For some tests where reduced speed is needed there will be a NPRA vehicle in front and at the back of motorcade. Communication to the vehicles will be via FM radio.

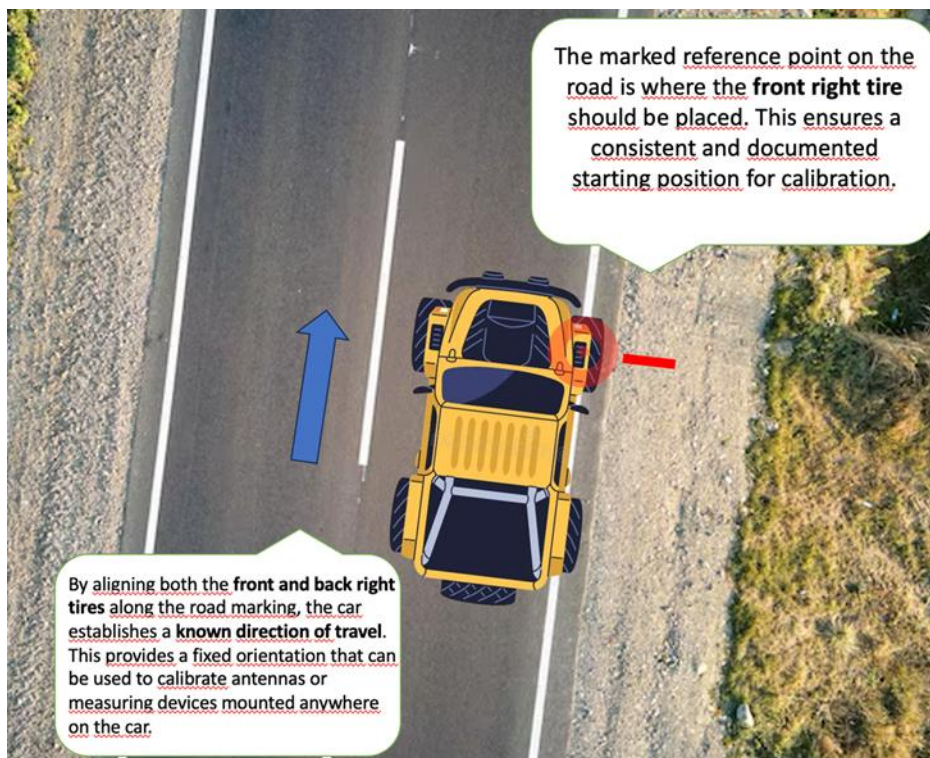
Calibration Control Marks (New at Jammertest 2025)

In a straight, open road section, three pairs of control marks (301-306) will be painted in center of the dotted white line at the right-hand side. The marks will be positioned so that vehicles can align with them as follows:

- The **front right wheel** is placed directly on the painted mark.
- The **rear right wheel** is aligned with the center of the white line.

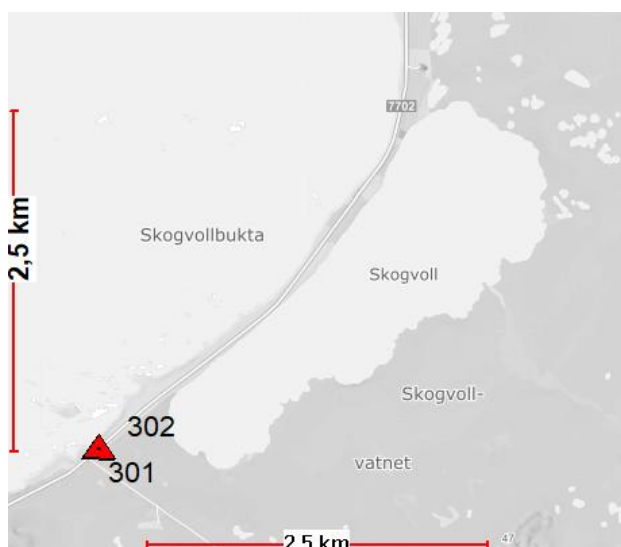
This arrangement provides a precise and repeatable reference for the forward right wheel position relative to the vehicle's facing direction.

The coordinates of the control marks will be averaged from several time independent RTK-measurements and distributed in *Appendix A Table 1*. These marks are intended for calibrating on-board GNSS and inertial navigation equipment against a known reference point.





JAM-S3 is the Jammer-point at the car roof on Site 3



Example, one pair of calibration points

